



C#

Level 1

Common Language
Specification (CLS)



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Definition

 CLS — Common Language Specification

 CLS is a set of rules that .NET languages follow to improve interoperability between code written in different .NET languages.

 It represents a subset of the Common Type System rules intended to support language interoperability.

Explanation

Different Languages

-  .NET supports multiple languages such as C#, Visual Basic, and F#.
-  Each language has its own syntax and language-specific features.

Common Rules

-  CLS defines a common set of rules for public APIs.
-  Following CLS rules makes libraries easier to consume from other CLS-compliant .NET languages.

Language Interoperability

C#



.NET



VB

-  C# code can interact with VB code when compatible APIs are used.
-  Libraries can be shared between .NET languages.

Procedures

1. Write code in a .NET language.
2. Define public types and members.
3. Follow CLS rules for public APIs when cross-language interoperability is required.
4. Compile the code into .NET assemblies.
5. Allow other compatible .NET languages to consume the library.

💡 Example

C# Library



CLS-compliant Public API



.NET Assembly



Visual Basic / F# / Other .NET Language

➡ Result: Different .NET languages can work with the same library when its exposed API follows compatible rules.

Note

- ◆ CLS is a specification/set of rules, not a runtime service like the Garbage Collector or JIT.
- ◆ CLS focuses mainly on language interoperability.
- ◆ Not every feature available in one .NET language is necessarily CLS-compliant for public APIs.

⊘ Warning

◆ Do not confuse CLS with CTS.

▨ CTS

- ✓ Defines the common .NET type system
- ✓ Describes the types available to .NET languages

▤ CLS

- ✓ Defines rules for language interoperability
- ✓ Represents a subset of CTS-related rules for public APIs



Comparison

▨ CTS

- ✓ Defines .NET types
- ✓ Broader type system
- ✓ Focuses on type compatibility
- ✓ Provides the foundation for .NET language interoperability

▨ CLS

- ✓ Defines common language rules
- ✓ Focuses on language interoperability
- ✓ Helps different .NET languages consume compatible APIs
- ✓ Represents a subset of CTS rules

✦ Summary

- ✓ CLS = Common Language Specification
- ✓ CLS defines rules for interoperability between .NET languages
- ✓ CTS defines the common .NET type system
- ✓ CLS represents a subset of CTS-related rules
- ✓ CLS is especially important for public APIs intended for use across .NET languages
- ✓ CLS is a specification, not a runtime component like JIT or GC

Thank you 😊